



Freelancer Management SaaS Application

Web Development Domain

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INTRODUCTION

The Problem We Solve



Before

- ✗ Excel for finances
- ✗ Trello for tasks
- ✗ Word for invoices
- ✗ Multiple apps for business
- ✗ Chaotic workflow



After

- ✓ Single dashboard
- ✓ All-in-one solution
- ✓ Professional invoices
- ✓ Time & project tracking
- ✓ Streamlined workflow

OBJECTIVES

Project Objectives



Client CRM

Manage clients with full CRUD operations



Time Tracking

Real-time stopwatch & manual entry



Invoicing

Generate invoices with PDF download



Reports

Financial charts & analytics



Security

Data isolation & JWT auth



Responsive UI

Works on all devices

FEATURES

Key Features Implemented

Pages

**API
Routes**

Models

Responsive



Contacts

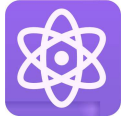


Proposals



Contracts

Technology Stack



React.js
Frontend



Node.js
Backend



MongoDB
Database



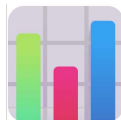
Express
API Framework



TypeScript
Language



CSS3
Styling



Recharts
Charts



jsPDF
PDF Gen



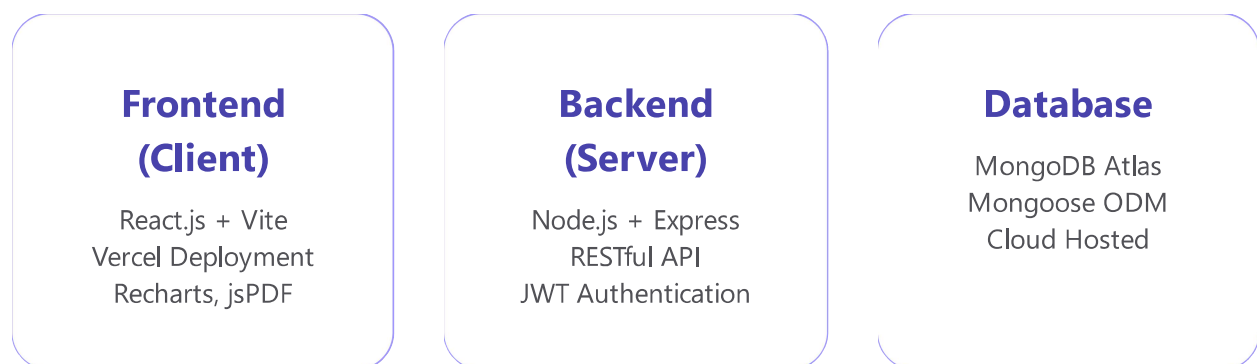
JWT
Auth



Vercel

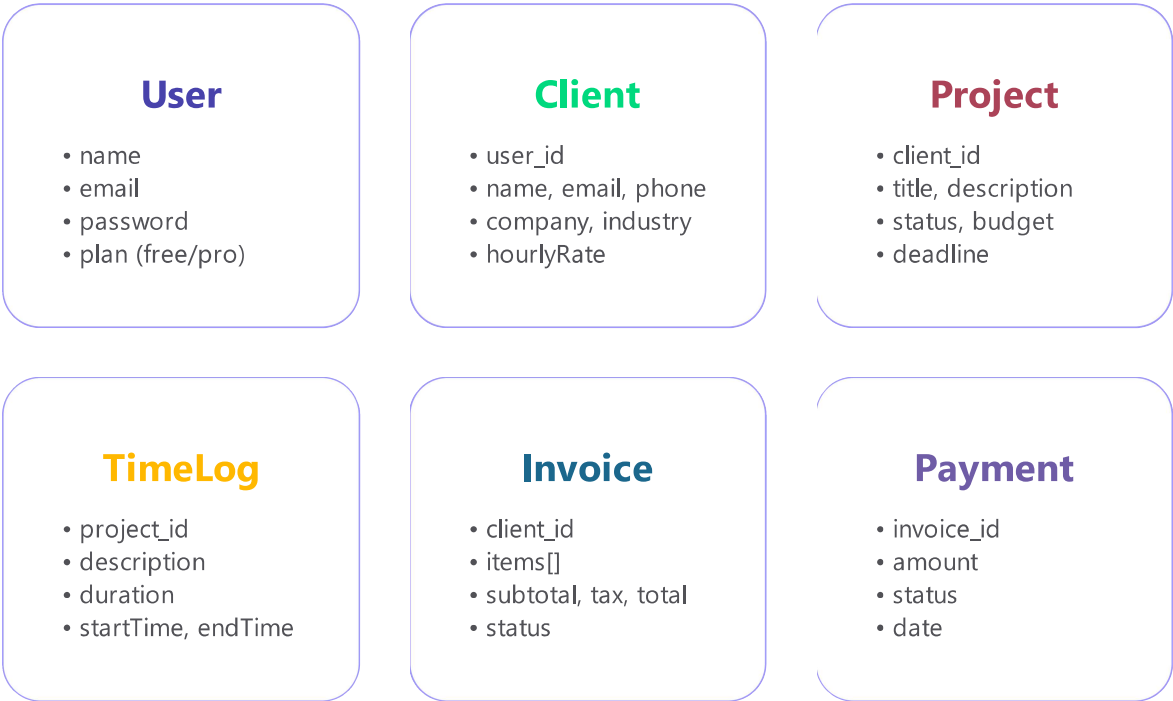
ARCHITECTURE

System Architecture



DATABASE

Database Schema



LOGIC

Mathematical Logic

Invoice Calculation

```
// Calculate Invoice Total
const subtotal =
items.reduce((sum, item) =>
  sum + (item.hours * item.rate),
0);

taxAmount = subtotal * (taxRate /
100);
total = subtotal + taxAmount;
```

Time × Rate

10 hrs ×
₹100 =
₹1000

Burn Rate

50 hrs ×
₹85 =
₹4250
spent

SECURITY

Security Implementation



Data Isolation

Every query includes
user_id



JWT Tokens

Access (15min) +
Refresh (7days)



Password Hashing

bcrypt with 12 salt
rounds



CORS Protection

Configured for
production



Rate Limiting

Prevents brute force



Tiered Access

Free (2 clients) / Pro
(unlimited)

```
// Data Isolation Example
const clients = await Client.find({ user: req.user.id });
```

DEVELOPMENT

Development Timeline

Week 1: CRM Core

- ✓
- Authentication System
- ✓ Client CRUD
- ✓ Project CRUD
- ✓ Task Management
- ✓ Data Isolation

Week 2: Time Tracking

- ✓
- Stopwatch Component
- ✓ Manual Time Entry
- ✓ Time Log Backend
- ✓ Budget Calculations
- ✓ Hours per Project

Week 3: Invoicing

- ✓ Invoice Creation Wizard
- ✓ PDF Generation (jsPDF)
- ✓ Mark as Billed
- ✓ Payment Tracking
- ✓ GST Calculations

Week 4: Polish

- ✓ Financial Charts
- ✓ Reports Page
- ✓
- Dashboard Stats
- ✓
- Responsive Design
- ✓ Cloud Deployment

CHALLENGES

Challenges Faced & Solutions



Timer Persistence

Solution:
Used localStorage to persist timer across page navigation



PDF Generation

Solution:
Implemented client-side PDF using jsPDF library



Data Isolation

Solution:
Scoped all queries to req.user.id middleware



CORS Issues

Solution:
Configured proper CORS headers for production



Typing in Forms

Solution:
Used React Portals for modals, removed double state updates



Responsive Design

Solution:
CSS Grid + Flexbox with media queries

SCREENSHOTS

Application Screenshots



Dashboard

Overview stats,
charts, quick
actions



Clients Page

Grid/List view,
add/edit/delete



Invoices

Line items,
calculations, PDF
download



Time Tracking

Stopwatch, manual
entry, logs



Reports

Revenue charts,
expense
breakdown



Projects

Status tracking,
budget
management

LIVE DEMO

Live URLs

**Frontend (Vercel)**

freelanceflow-blue-delta.vercel.app

**Backend API (Render)**

freelanceflow-api-80zc.onrender.com



Quick Test Guide

1. Open the frontend URL
2. Register a new account
3. Click "Load Sample Data" on Dashboard
4. Explore Clients, Projects, Invoices
5. Download an Invoice as PDF
6. View Reports with Charts



Questions & Discussion

🚀 Full-Stack MERN Application | Production Deployed | Ready to Use

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 GitHub: github.com/farhankh8/freelanceflow